

Khulna University of Engineering & Technology
Department of Computer Science and Engineering

CSE 4222: Natural Language Processing Laboratory

Project Report on
A Lightweight Bangla Review Analytics
Platform for E-Commerce Sellers

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1 Introduction and Project Overview

Online customer feedback on e-commerce platforms provides crucial intelligence for consumers making purchasing decisions and merchants seeking operational refinement. However, authentic customer reviews rarely convey a single, monolithic sentiment. On platforms such as Daraz Bangladesh, a customer review frequently commends the core product while simultaneously voicing frustration regarding logistics or damaged packaging (e.g., “*The headphone is authentic and works well, but delivery took two weeks and packaging was torn*”).

Traditional document-level sentiment classification flattens these multi-dimensional sentiments into a single polarity score (positive, negative, or neutral), obscuring actionable aspect-specific insights. **Aspect-Based Sentiment Analysis (ABSA)** addresses this limitation by decomposing opinion text into distinct aspect categories (e.g., *Product Quality*, *Price*, *Delivery*, *Packaging*, *Seller Service*) and determining their respective polarities independently.

1.1 Linguistic Challenges in Bangla ABSA

Executing fine-grained ABSA on the Bangla language presents unique linguistic and computational challenges:

1. **Morphological Variations and Orthographic Noise:** Bangla text exhibits high morphological richness, frequent colloquialisms, informal phonetic spellings, repeated character elongation for emphasis (e.g., *onek* → *oneeeek*), and invisible zero-width Unicode characters (ZWNJ, ZWJ, BOM).
2. **Code-Mixing and Loanwords:** E-commerce reviews routinely mix English brand names, technical terms, and Latin characters alongside native Bangla script, requiring cleaning routines that do not discard valid alphanumeric tokens.
3. **Severe Aspect Imbalance:** The corpus exhibits pronounced class skew: *Product Quality* appears in 98.2% of reviews, whereas *Packaging* (3.7%) and *Seller Service* (5.5%) represent rare, long-tail categories where data scarcity impedes model generalization.

1.2 Project Objectives

This project develops an end-to-end, modular NLP pipeline for hierarchical ABSA on Bangla Daraz customer reviews with the following core deliverables:

- A structured **Hierarchical 3-Level Task Formulation** (Global Sentiment → Multi-Label Aspect Detection → Conditioned Aspect Polarity).
- A deterministic **5-Stage Specialized Bangla Preprocessing Pipeline** integrating Unicode NFC canonicalization, zero-width mark stripping, noise cleaning, de-elongation, and token/negation preservation.
- A systematic **Empirical Benchmark** comparing a Subword Hybrid TF-IDF linear pipeline with a Deep 2-layer Bidirectional LSTM (BiLSTM) network across identical stratified test splits.

- A production-ready **Streamlit Analytical Dashboard** supporting model family switching, real-time single-review ABSA diagnostics, confidence meters, batch CSV processing, and live benchmark comparisons.

2 System Architecture and Workflow

The end-to-end system follows a modular hierarchical architecture that processes raw Bangla text through specialized linguistic cleaning, extracts numerical feature representations, and routes predictions through a 3-tier classification hierarchy, as illustrated in Figure 1.

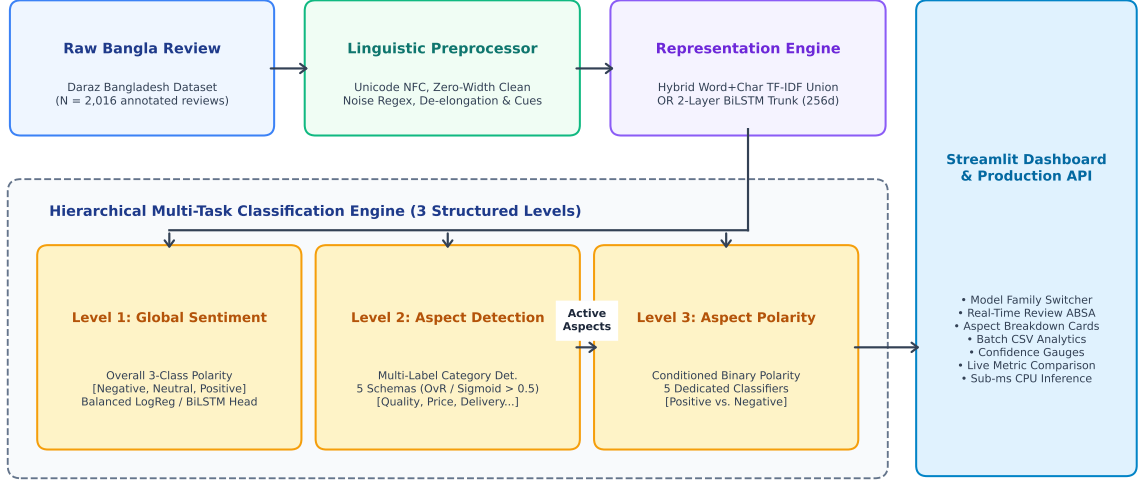


Figure 1: End-to-End Hierarchical ABSA System Architecture and Workflow.

2.1 Hierarchical 3-Level Decision Pipeline

Let an input Bangla review be a token sequence $T = (w_1, w_2, \dots, w_m)$. The inference pipeline executes across three structured tiers:

- **Level 1: Global Sentiment Classification.** Predict the overall document polarity $y_{\text{sent}} \in \{\text{Negative}, \text{Neutral}, \text{Positive}\}$.
- **Level 2: Multi-Label Aspect Category Detection.** Identify all aspect categories discussed in review T from a schema of $K = 5$ core dimensions:

$$\mathcal{A} = \{\text{Product Quality, Price, Delivery, Packaging, Seller Service}\} \quad (1)$$

The target is formalized as a binary indicator vector $\mathbf{a} = [a_1, a_2, \dots, a_K]^T \in \{0, 1\}^K$ evaluated against a decision threshold $\tau = 0.5$.

- **Level 3: Aspect-Specific Polarity Classification.** For each active aspect category k satisfying $a_k = 1$, predict its specific polarity $p_k \in \{\text{Negative}, \text{Positive}\}$. Aspects not detected in Level 2 are skipped, ensuring computational efficiency and eliminating false-positive polarity predictions.

3 Specialized Bangla Preprocessing Methodology

To handle informal orthography and noisy user feedback in e-commerce reviews, we construct a deterministic, 5-stage preprocessing pipeline implemented in *src/preprocessing.py*, shown in Figure 2.

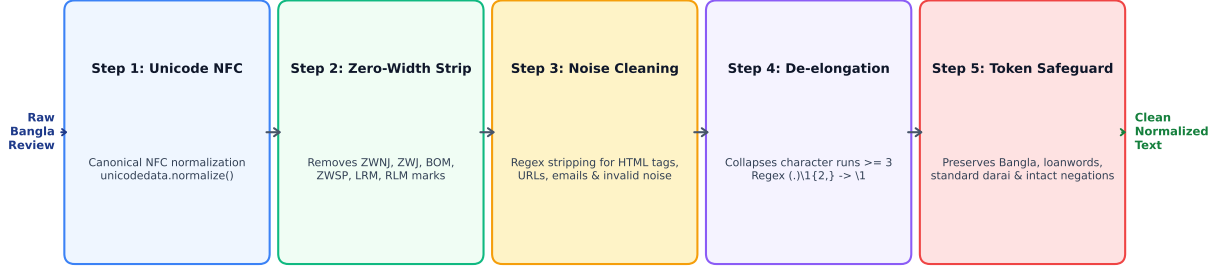


Figure 2: The 5-Stage Specialized Bangla Linguistic Preprocessing Pipeline.

3.1 Step-by-Step Preprocessing Mechanisms

1. **Unicode NFC Normalization:** Text is canonicalized into Unicode Normalization Form C (NFC) via *unicodedata.normalize*, resolving disparate Unicode encodings for composite graphemes and vowel diacritics.
2. **Zero-Width Character Stripping:** Invisible artifacts including Zero-Width Non-Joiners (ZWNJ, *U+200C*), Zero-Width Joiners (ZWJ, *U+200D*), Byte Order Marks (*U+FEFF*), Zero-Width Spaces (*U+200B*), and directional marks (*U+200E*, *U+200F*) are stripped using fast string translation tables.
3. **Noise and Web Artifact Cleansing:** Pre-compiled regular expressions eliminate HTML markup (*<[^\>]+>*), HTTP/HTTPS URLs, email addresses, and non-standard symbols.
4. **Character Run De-elongation:** Expressive vowel or consonant prolongations of length ≥ 3 are collapsed to a single character using the regular expression *(.)\1{2,} → \1* (e.g., “*khuuub*” → “*khub*”, “*onekbbb*” → “*onekb*”), curbing vocabulary explosion.
5. **Token and Negation Safeguards:** The pipeline explicitly preserves the Bangla Unicode script range (*\u0980–\u09FF*), English alphanumeric tokens (*A–Za–z0–9*) for code-mixed brand and model names, Bengali punctuation (*darai* —), and critical polarity negation cues (*na*, *nai*, *nei*, *noy*, *kintu*) to prevent polarity inversion errors during downstream vectorization.

4 Model Architectures and Methodologies

We benchmark two complementary modeling paradigms: a subword-enhanced linear pipeline and a deep recurrent neural network.

4.1 Pipeline 1: Hybrid TF-IDF Feature Union + Balanced Linear Models

To capture word-level semantics and sub-morphemic lexical patterns, we construct a heterogeneous feature union $\mathbf{x}_{\text{tfidf}} \in \mathbf{R}^D$ (*src/features.py*):

$$\mathbf{x}_{\text{tfidf}} = [\mathbf{x}_{\text{word}} \parallel \mathbf{x}_{\text{char}}] \quad (2)$$

where:

- $\mathbf{x}_{\text{word}}$ extracts word n -grams with range $(1, 2)$ matching token pattern `[\u0980-\u09FFA-Za-z0-9]` + with document frequency bounds $\text{df} \in [2, 0.95 \cdot N]$.
- $\mathbf{x}_{\text{char}}$ extracts character n -grams within word boundaries (*char_wb*) with range $(3, 5)$ to capture morphological affixes and stems.
- Sublinear term frequency scaling is applied: $\text{tf}_{\text{sub}} = 1 + \log(\text{tf})$ for $\text{tf} > 0$.

For classification, we deploy L_2 -regularized Logistic Regression ($C = 1.0$) optimized via L-BFGS with inverse class-frequency weighting:

$$\min_{\mathbf{w}, b} \frac{1}{2} \|\mathbf{w}\|_2^2 + C \sum_{i=1}^N \alpha_{y_i} \ell(y_i, \mathbf{w}^T \mathbf{x}_i + b), \quad \alpha_c = \frac{N}{J \cdot N_c} \quad (3)$$

where J denotes the number of classes. For Level 2 multi-label aspect detection, a One-vs-Rest (OvR) scheme with $K = 5$ binary estimators is trained.

4.2 Pipeline 2: Deep Bidirectional LSTM (BiLSTM)

The deep sequential baseline processes token sequences through recurrent temporal representations, as shown in Figure 3.

The BiLSTM architecture (*LSTMBase* in *src/features.py*) comprises:

1. **Vocabulary and Embedding Layer:** Sequences are padded or truncated to length $L = 128$ over a vocabulary $|\mathcal{V}| = 15,000$ ($\text{min_freq} = 2$) and projected into dense embeddings $\mathbf{E} \in \mathbf{R}^{|\mathcal{V}| \times d}$ ($d = 300$, *padding_idx=0*), followed by spatial dropout ($p = 0.5$).
2. **Stacked BiLSTM Trunk:** A 2-layer BiLSTM network captures bidirectional temporal dependencies:

$$\vec{h}_t^{(l)} = \text{LSTM}_{\text{fwd}}^{(l)}(\mathbf{h}_t^{(l-1)}, \vec{h}_{t-1}^{(l)}) \quad (4)$$

$$\overleftarrow{h}_t^{(l)} = \text{LSTM}_{\text{bwd}}^{(l)}(\mathbf{h}_t^{(l-1)}, \overleftarrow{h}_{t+1}^{(l)}) \quad (5)$$

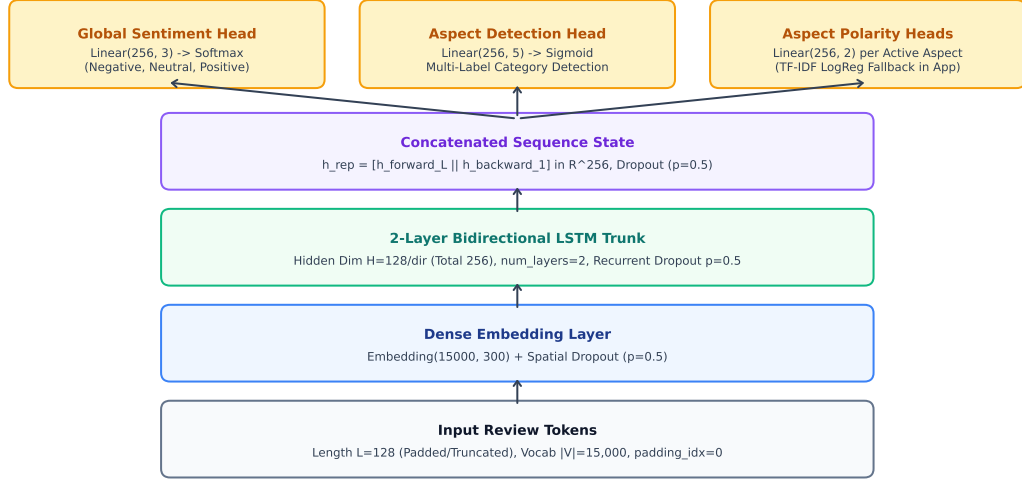


Figure 3: Deep Bidirectional LSTM (BiLSTM) Multi-Task Neural Network Architecture.

with hidden dimension $H = 128$ per direction (total 256 dimensions) and inter-layer dropout $p = 0.5$.

3. **Sequence Representation:** The sentence representation $\mathbf{h}_{\text{rep}} \in \mathbf{R}^{2H}$ concatenates the final forward and backward states: $\mathbf{h}_{\text{rep}} = [\bar{h}_L^{(2)} \parallel \overleftarrow{h}_1^{(2)}]$, followed by a dropout layer ($p = 0.5$).
4. **Task Classification Heads:**
 - *Global Sentiment Head (SentimentLSTM):* A linear layer projecting $256 \rightarrow 3$ outputs trained via Cross-Entropy loss.
 - *Aspect Detection Head (AspectLSTM):* A linear layer projecting $256 \rightarrow 5$ outputs trained via Binary Cross-Entropy with Logits (*BCEWithLogitsLoss*).
 - *Aspect Polarity Heads:* Binary linear heads projecting $256 \rightarrow 2$ outputs. In the production web application (*app.py*), aspect polarities leverage the tuned TF-IDF linear classifiers as the robust inference engine due to sample sparsity in low-frequency aspects.

Both PyTorch models are optimized using Adam ($\eta = 10^{-3}$, batch size $B = 32$, 10 epochs).

5 Experimental Results and Empirical Evaluation

5.1 Dataset Partitioning

The empirical benchmark was conducted on $N = 2,016$ labeled Bangla reviews from Daraz Bangladesh, partitioned using a stratified split (Seed = 42):

- **Training Set:** $N_{\text{train}} = 1,612$ (80%)
- **Held-Out Test Set:** $N_{\text{test}} = 404$ (20%)

5.2 Quantitative Performance Comparison

Table 1 presents the quantitative benchmark results recorded across all three hierarchical levels on the test partition.

Table 1: Comprehensive performance comparison across all hierarchical tasks on the test set ($N_{\text{test}} = 404$).

Task / Target	Architecture	Accuracy	Macro-F1	Weighted-F1	Diagnostic Metric
Global Sentiment	TF-IDF + LogReg	0.9158	0.8490	0.9197	–
	PyTorch BiLSTM	0.8886	0.7846	0.8874	–
Aspect Detection	TF-IDF + OvR	0.9644	0.8038	0.9376	Micro-F1: 0.9364
	PyTorch BiLSTM	0.9609	0.4986	0.9048	Micro-F1: 0.9281
Aspect Polarity Classification:					
<i>Product Quality</i>	TF-IDF + LogReg	0.9725	0.9514	0.9729	$N_{\text{train}} = 1,452$
	PyTorch BiLSTM	0.9615	0.9292	0.9613	$N_{\text{train}} = 1,452$
<i>Price</i>	TF-IDF + LogReg	0.9789	0.7446	0.9738	$N_{\text{train}} = 376$
	PyTorch BiLSTM	0.9474	0.6292	0.9510	$N_{\text{train}} = 376$
<i>Delivery</i>	TF-IDF + LogReg	0.8772	0.7580	0.8683	$N_{\text{train}} = 225$
	PyTorch BiLSTM	0.8070	0.5778	0.7797	$N_{\text{train}} = 225$
<i>Packaging</i>	TF-IDF + LogReg	0.8000	0.7205	0.7702	$N_{\text{train}} = 57$
	PyTorch BiLSTM	0.6667	0.4000	0.5333	$N_{\text{train}} = 57$
<i>Seller Service</i>	TF-IDF + LogReg	0.9545	0.4884	0.9323	$N_{\text{train}} = 86$
	PyTorch BiLSTM	0.9545	0.4884	0.9323	$N_{\text{train}} = 86$

5.3 Key Findings and Discussion

1. **Superiority of Subword Character n -Grams:** The hybrid TF-IDF linear pipeline outperforms the BiLSTM architecture across all tasks, attaining higher global sentiment accuracy (91.58% vs. 88.86%) and superior sentiment Macro-F1 (0.8490 vs. 0.7846). The character n -gram feature union ($3 \leq n \leq 5$) directly captures sub-morphemic roots and negation affixes without requiring large pretraining corpora.
2. **Data Scarcity in Neural Sequential Models:** The BiLSTM architecture, training embeddings from scratch on $N_{\text{train}} = 1,612$ samples, exhibits diminished recall on rare categories (aspect detection Macro-F1 of 0.4986 vs. 0.8038 for TF-IDF). Its unconstrained parameter space suffers from sample scarcity compared to class-weighted convex linear models.
3. **Aspect Polarity Dynamics:** On high-support aspects such as *Product Quality* ($N_{\text{train}} = 1,452$), both models demonstrate high accuracy (97.25% vs. 96.15%). On rare aspects such as *Packaging* ($N_{\text{train}} = 57$), TF-IDF achieves substantially higher Macro-F1 (0.7205 vs. 0.4000).
4. **Computational Efficiency:** The TF-IDF linear pipeline delivers sub-millisecond CPU inference per review, making it optimal for lightweight, production web deployment.

6 System Implementation and Deployment

The codebase is structured into modular Python components with clear separation of responsibilities:

- *src/config.py*: Central configuration constants, hyperparameter definitions, and path mappings.
- *src/preprocessing.py*: High-throughput 5-stage Unicode normalization, regex de-elongation, and token safeguards.
- *src/data_processing.py*: Dataset ingestion, multi-delimiter label parsing (# and ;), and stratified splitting.
- *src/features.py*: Heterogeneous TF-IDF FeatureUnion and PyTorch *BanglaVocab* and *LSTMBase* networks.
- *src/train.py* & *src/evaluate.py*: Automated multi-tier model training, evaluation metrics, and benchmark exports.
- *src/persistence.py*: Serialization and deserialization for TF-IDF (.pkl) and PyTorch (.pt) models.
- *src/predict.py*: Unified inference orchestrator bundling preprocessing and hierarchical ABSA inference.
- *app.py*: Production Streamlit dashboard with model switching, real-time diagnostics, and batch CSV analysis.

7 Conclusion

This project presented an end-to-end, methodology-focused realization of Hierarchical Aspect-Based Sentiment Analysis on Bangla e-commerce customer reviews. By combining domain-tailored linguistic preprocessing (Unicode canonicalization, zero-width stripping, character de-elongation, and negation safeguards) with a hybrid TF-IDF feature union and class-weighted linear models, our system achieves 91.58% sentiment accuracy and a 0.9364 aspect Micro-F1. The resulting modular pipeline and interactive Streamlit web platform deliver efficient, actionable review analytics for e-commerce merchants.